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REPORT DOCUMENTATION PAGE (SF298)
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Hierarchical reconstruction with up to second degree remainder for solving nonlinear conservation laws*

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Abstract

The hierarchical reconstruction (HR) (Liu *et al* 2007 *SIAM J. Numer. Anal.* **45** 2442–67) can effectively reduce spurious oscillations without local characteristic decomposition for numerical capturing of discontinuous solutions. However, there are still small remaining overshoots/undershoots in the vicinity of discontinuities. HR with partial neighbouring cells (Xu *et al* 2009 *J. Comput. Phys.* **228** 2194–212) essentially overcomes this drawback for the third order case, and in the mean time further improves the resolution of the numerical solution. Extending the technique to higher order cases we observe the returning of overshoots/undershoots. In this paper, we introduce a new technique to work with HR on partial neighbouring cells, which lowers the order of the remainder of the polynomial in the current cell while maintaining the theoretical order of accuracy, essentially eliminates overshoots/undershoots for the fourth and fifth order cases and reduces the numerical cost.

1. Introduction

Weak solutions of nonlinear conservation laws contain discontinuities, which provide an interesting subject for numerical study and motivate many fundamental numerical techniques, for example the Godunov scheme [7], MUSCL scheme [25–27], TVD scheme [8], ENO [9, 23, 24] and WENO schemes [11, 13] and many others. As the formal order of accuracy of a finite volume scheme becomes higher, local characteristic decomposition is usually required. For instance in [17], numerical experiments show that without local characteristic

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decomposition, spurious oscillations start to show up for both the upwind WENO scheme and a central scheme with WENO reconstruction when the order is higher than three, because the ENO (or WENO) reconstruction locally selects a stencil of cells within the smooth part of the solution to reconstruct a high order polynomial, which is only possible if the solution is decomposed in terms of local characteristic variables in the vicinities of interacting discontinuities. If the numerical solution is represented as a piecewise polynomial, e.g. in the discontinuous Galerkin (DG) method [4–6, 19], or in a finite volume scheme with a preliminary reconstruction (to generate a piecewise polynomial solution), the WENO strategy can be applied to the cell averages of these polynomials [18]. Compact limiting techniques which are supposed to remove spurious oscillations using information only from adjacent cells for any orders include the TVD/TVB limiter [6], the moment limiter [2] and the recently developed hierarchical reconstruction (HR) [15].

HR decomposes the job of limiting a high order polynomial defined in a cell (which may contain spurious oscillations) into a series of smaller jobs, each of which involves only the non-oscillatory reconstruction of a linear polynomial, e.g. the MUSCL reconstruction. Therefore it uses only information from adjacent cells and can be formulated on unstructured meshes in multi-dimensions. It does not use local characteristic decomposition and thus is less dependent on the underlying equation to be solved. Nevertheless in [15, 16], small overshoots/undershoots after interactions of discontinuities can still be observed. HR first takes certain derivative for the polynomials defined in the current cell and its adjacent cells as in the moment limiter. Then it takes a reconstruction approach, namely it estimates cell averages of the linear part of the polynomial in the current cell over adjacent cells to certain order of accuracy, and applies a non-oscillatory reconstruction to recompute the linear part. The remainder of the polynomial in the current cell (after removing the linear part) plays a key role in estimating cell averages (of the linear part) over adjacent cells. If we estimate only cell averages over partial neighbouring cells [30], namely the fraction of adjacent cells (usually half in size) which are closer to the current cell, the remainder in the current cell can be extended over a shorter distance in HR which essentially eliminates overshoots/undershoots in the non-smooth part of the solution in the third order case. However, when the degree of the remainder gets higher (in higher order cases), we may not be able to estimate cell averages of the remainder over smaller partial neighbouring cells since it may tolerate more artefacts in the vicinities of interacting discontinuities. This leads us to think of lowering the degree of the remainder in higher order cases. The consideration is to do so without hurting the provable order of accuracy or introducing numerical instability. We would also like to mention that the spectral volume method [28] seems to be more tolerating for limiting techniques. In [31], HR (without using partial neighbouring cells) has been successfully applied to the third order spectral volume method on triangular meshes.

2. A brief review of the HR

Here we review the HR procedure introduced in [15]. Suppose that the computational domain $\Omega \in \mathbb{R}^d$ is a region associated with a mesh $\{C_I : I = 1, 2, \dots, N\}$, where C_I is called a cell which is a bounded open set with piecewise smooth boundary, x_I is the centroid of C_I and $\bigcup_{I=1}^N \overline{C_I} = \overline{\Omega}$. Let Δx be the maximum of the diameters of C_I for all I . Suppose the numerical solution is represented in each cell C_I by a polynomial $U_I(x - x_I)$ of degree r , though it may contain spurious oscillations. The HR procedure is to recompute the polynomial $U_I(x - x_I)$ by using polynomials in cells adjacent to C_I . These adjacent cells are collected as the set $\{C_J\}$ (which also contains cell C_I) and the polynomials (of degree r) supported on them

are thus renamed as $\{U_J(x - x_J)\}$, respectively. We need to recompute the new coefficients

$$\frac{1}{m!} \tilde{U}_I^{(m)}(\mathbf{0}), \quad |\mathbf{m}| = r, r-1, \dots, 0$$

for $U_I(x - x_I)$ (written in terms of its Taylor expansion around x_I) iteratively from the highest to the lowest degree terms.

To obtain $\tilde{U}_I^{(m)}(\mathbf{0})$, we first compute many *candidates* of $U_I^{(m)}(\mathbf{0})$ (sometimes still denoted as $\tilde{U}_I^{(m)}(\mathbf{0})$ with specification), and we then let the new value for $U_I^{(m)}(\mathbf{0})$ be

$$\tilde{U}_I^{(m)}(\mathbf{0}) = F(\text{candidates of } U_I^{(m)}(\mathbf{0})),$$

where F is a convex limiter of its arguments (e.g., the minmod function to be specified later), $F(a_1, a_2, \dots, a_l) = \sum_{i=1}^l \theta_i a_i$, for some $\theta_i \geq 0$ and $\sum_{i=1}^l \theta_i = 1$.

In order to find these candidates of $U_I^{(m)}(\mathbf{0})$, $|\mathbf{m}| = m$, we take a $(m-1)$ th order partial derivative of $U_I(x - x_I)$ (and also polynomials in adjacent cells), and express

$$\partial^{m-1} U_I(x - x_I) = L_I(x - x_I) + R_I(x - x_I),$$

where L_I is the linear part (containing the zeroth and first degree terms) and R_I is the remainder. Clearly, every coefficient in the first degree terms of L_I is in the set $\{U_I^{(m)}(\mathbf{0}) : |\mathbf{m}| = m\}$. And for every $\mathbf{m}$ subject to $|\mathbf{m}| = m$, one can always take some $(m-1)$ th order partial derivatives of $U_I(x - x_I)$ so that $U_I^{(m)}(\mathbf{0})$ is a coefficient in a first degree term of L_I . Thus, a ‘candidate’ for a coefficient in a first degree term of L_I is the candidate for the corresponding $U_I^{(m)}(\mathbf{0})$.

In order to find a set of candidates for all coefficients in the first degree terms of $L_I(x - x_I)$, we only need to know the new approximate cell averages of $L_I(x - x_I)$ on $d+1$ distinct mesh cells adjacent to cell C_I . Assume $C_{J_0}, C_{J_1}, \dots, C_{J_d} \in \{C_J\}$ are these cells and $\bar{L}_{J_0}, \bar{L}_{J_1}, \dots, \bar{L}_{J_d}$ are the corresponding new approximate cell averages. The set of these $d+1$ cells with the associated cell averages is called a *stencil*. Let a linear polynomial $\tilde{L}_I(x - x_I)$ be determined by the linear system

$$\frac{1}{|C_{J_l}|} \int_{C_{J_l}} \tilde{L}_I(x - x_I) dx = \bar{L}_{J_l}, \quad l = 0, 1, \dots, d. \quad (2.1)$$

Then the coefficients in the first degree terms of $\tilde{L}_I(x - x_I)$ become the candidates for the corresponding coefficients of $L_I(x - x_I)$. Therefore, a stencil located near cell C_I will determine a set of candidates for all coefficients in the first degree terms of $L_I(x - x_I)$. The key is to compute the new approximate cell averages of $L_I(x - x_I)$ on the cells of $\{C_J\}$, which is outlined by the following algorithm.

Algorithm 1.

Step 1. Suppose $r \geq 2$. For $m = r, r-1, \dots, 1$, do the following:

- Take a $(m-1)$ th order partial derivative for each of $\{U_J(x - x_J)\}$ to obtain polynomials $\{\partial^{m-1} U_J(x - x_J)\}$, respectively. In particular, denote $\partial^{m-1} U_I(x - x_I) = L_I(x - x_I) + R_I(x - x_I)$, where $L_I(x - x_I)$ is the linear part of $\partial^{m-1} U_I(x - x_I)$ and $R_I(x - x_I)$ is the remainder.
- For all J , calculate the cell average of $\partial^{m-1} U_J(x - x_J)$ on cell C_J to obtain $\overline{\partial^{m-1} U_J}$.
- Let $\tilde{R}_I(x - x_I)$ be the $R_I(x - x_I)$ with its coefficients replaced by the corresponding new values⁴. For all J , calculate the cell averages of $\tilde{R}_I(x - x_I)$ on the cell C_J to obtain $\bar{R}_J$.
- Let $\bar{L}_J = \overline{\partial^{m-1} U_J} - \bar{R}_J$ for all J .

⁴ At this stage, we have already found new values for all coefficients in the terms of $U_I(x - x_I)$ of degree above m . These coefficients remain in $R_I(x - x_I)$ (after taking a $(m-1)$ th order partial of $U_I(x - x_I)$). When they are replaced by their corresponding new values, $R_I(x - x_I)$ becomes $\tilde{R}_I(x - x_I)$.

- (e) Form stencils out of the new approximate cell averages $\{\bar{L}_J\}$ by using a MUSCL, second order ENO or other non-oscillatory strategies. Each stencil will determine a set of candidates for the coefficients in the first degree terms of $L_I(\mathbf{x} - \mathbf{x}_I)$, which are also candidates for the corresponding $U_I^{(m)}(\mathbf{0})$'s, $|\mathbf{m}| = m$.
- (f) Repeat from (a) to (e) until all possible combinations of the $(m - 1)$ th order partial derivatives are taken. Then the candidates for all coefficients in the m th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$ have been computed. For each of these coefficients, say $\frac{1}{m!}U_I^{(m)}(\mathbf{0})$, $|\mathbf{m}| = m$, let the new value $\tilde{U}_I^{(m)}(\mathbf{0}) = F(\text{candidates of } U_I^{(m)}(\mathbf{0}))$.

Step 2. The new coefficient in the zeroth degree term of $U_I(\mathbf{x} - \mathbf{x}_I)$ is chosen so that the cell average of $U_I(\mathbf{x} - \mathbf{x}_I)$ on cell C_I is invariant with the new coefficients. At this stage all the new coefficients of $U_I(\mathbf{x} - \mathbf{x}_I)$ have been found.

Even though the non-oscillatory reconstruction of linear polynomials used in algorithm 1 is only second order accurate (such as the MUSCL or second order ENO), the approximation order of accuracy of a polynomial in a cell is unaffected by the algorithm, because in step 1(d) the new approximate cell averages satisfy

$$\bar{L}_J = \frac{1}{|C_J|} \int_{C_J} L_I(\mathbf{x} - \mathbf{x}_I) d\mathbf{x} + \mathcal{O}((\Delta x)^{r-m+2}) \quad (2.2)$$

if $\{U_J(\mathbf{x} - \mathbf{x}_J)\}$ have optimal order of approximation to the solution which is locally smooth enough.

3. A technique to lower the degree of the remainder

Note that in step 1(c) of algorithm 1, the updated remainder $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ needs to be averaged outside the cell C_I , which is not a problem in smooth regions of the solution. However, when the involved polynomials contain spurious oscillations, it could cause small overshoots/undershoots from our numerical experience. In the third order case (i.e., $r = 2$ and $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ can be up to second degree), this problem has been solved in [30] by applying HR with partial neighbouring cells (about half of the original size) of C_I so that they are closer to C_I . In higher order cases, $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ can be above second degree and further bringing partial neighbouring cells close to C_I could introduce artefacts near interacting discontinuities of the solution. Simply truncating $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ to up to second degree will hurt the accuracy. Our new technique uses only second degree terms of $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ in the estimation of $\bar{L}_J$ when $J \neq I$, and should work with partial neighbouring cells for $r \geq 3$.

Algorithm 2.

Step 1. Suppose $r \geq 3$. For $m = r, r - 1, \dots, 1$, do the following:

- (a) Take a $(m - 1)$ th order partial derivative for each of $\{U_J(\mathbf{x} - \mathbf{x}_J)\}$ to obtain polynomials $\{\partial^{m-1}U_J(\mathbf{x} - \mathbf{x}_J)\}$, respectively. In particular, denote $\partial^{m-1}U_I(\mathbf{x} - \mathbf{x}_I) = L_I(\mathbf{x} - \mathbf{x}_I) + R_I(\mathbf{x} - \mathbf{x}_I) = L_I(\mathbf{x} - \mathbf{x}_I) + Q_I(\mathbf{x} - \mathbf{x}_I) + \dots$, where $L_I(\mathbf{x} - \mathbf{x}_I)$ is the linear part of $\partial^{m-1}U_I(\mathbf{x} - \mathbf{x}_I)$, $R_I(\mathbf{x} - \mathbf{x}_I)$ is the remainder and $Q_I(\mathbf{x} - \mathbf{x}_I)$ is the second degree part of the remainder.
- (b) Let $\overline{\partial^{m-1}U_I}$ be the cell average of $\partial^{m-1}U_I$ over the cell C_I . For all $J \neq I$, rewrite $\partial^{m-1}U_J(\mathbf{x} - \mathbf{x}_J) = V_J(\mathbf{x} - \mathbf{x}_I) = L_J(\mathbf{x} - \mathbf{x}_I) + Q_J(\mathbf{x} - \mathbf{x}_I) + \dots$, where $L_J(\mathbf{x} - \mathbf{x}_I)$ is the linear part of $V_J(\mathbf{x} - \mathbf{x}_I)$ and $Q_J(\mathbf{x} - \mathbf{x}_I)$ is the second degree part; calculate the cell average of $L_J(\mathbf{x} - \mathbf{x}_I) + Q_J(\mathbf{x} - \mathbf{x}_I)$ on cell C_J to obtain $\overline{\partial^{m-1}U_J}$.

- (c) Let $\bar{R}_I$ be the cell average of $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ over cell C_I , where $\tilde{R}_I(\mathbf{x} - \mathbf{x}_I)$ is the $R_I(\mathbf{x} - \mathbf{x}_I)$ with its coefficients replaced by the corresponding new values, similarly for $\tilde{Q}_I(\mathbf{x} - \mathbf{x}_I)$. For all $J \neq I$, calculate the cell averages of $\tilde{Q}_I(\mathbf{x} - \mathbf{x}_I)$ on cell C_J to obtain $\bar{R}_J$.
- (d) Same as in algorithm 1.
- (e) Same as in algorithm 1.
- (f) Same as in algorithm 1.

Step 2. Same as in algorithm 1.

Remark. In algorithm 2, when $m = r$ and $r - 1$, the new coefficients are computed in exactly the same way as in algorithm 1 modulo the differences in steps 1(a)–(c).

Condition 1. Let $\{\mathbf{x}_{J_0}, \mathbf{x}_{J_1}, \dots, \mathbf{x}_{J_d}\}$ be the $d + 1$ cell centroids of a stencil. Then there is a point among them, say $\mathbf{x}_{J_0}$, such that the matrix $A = (1/\Delta x)[\mathbf{x}_{J_1} - \mathbf{x}_{J_0}, \mathbf{x}_{J_2} - \mathbf{x}_{J_0}, \dots, \mathbf{x}_{J_d} - \mathbf{x}_{J_0}]$ is non-singular. Further, there is a constant $\beta > 0$ independent of Δx such that $\|A^{-1}\| \leq \beta$.

In 2D, this condition means that $\mathbf{x}_{J_0}, \mathbf{x}_{J_1}, \mathbf{x}_{J_2}$ are not along a straight line. Further, the angles of the triangle $\mathbf{x}_{J_0}, \mathbf{x}_{J_1}, \mathbf{x}_{J_2}$ have a positive lower bound independent of Δx .

Theorem 1. Suppose $\{U_J(\mathbf{x} - \mathbf{x}_J)\}$ in algorithm 2 approximate a C^{r+1} function $u(\mathbf{x})$ with point-wise error $\mathcal{O}((\Delta x)^{r+1})$ within their respective cell $\{C_J\}$, and all cells in $\{C_J\}$ are contained in a circle centred at $\mathbf{x}_I$ with radius $\mathcal{O}(\Delta x)$. Let the $d + 1$ cell centroids in every stencil used in algorithm 2 satisfy condition 1. Then after the application of algorithm 1, the polynomial $\tilde{U}_I(\mathbf{x} - \mathbf{x}_I)$, i.e. $U_I(\mathbf{x} - \mathbf{x}_I)$ with its coefficients replaced by the corresponding new values, also approximates the function $u(\mathbf{x})$ with point-wise error $\mathcal{O}((\Delta x)^{r+1})$ within cell C_I . The cell average of $\tilde{U}_I(\mathbf{x} - \mathbf{x}_I)$ on cell C_I is the same as that of $U_I(\mathbf{x} - \mathbf{x}_I)$.

Proof. The proof follows [15] exactly once we show that estimate (2.2) also holds for algorithm 2. From the assumption we know that the coefficients in the m th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$, $0 \leq m \leq r$, are the $(r - m + 1)$ th order approximation to the corresponding coefficients of the Taylor expansion of $u(\mathbf{x})$ at $\mathbf{x}_I$.

Assume that when starting to compute new values for the coefficients of the m th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$, $1 \leq m \leq r$, all the computed new values (if there are any) for the coefficients of the l th degree terms ($m < l \leq r$, if they exist) of $U_I(\mathbf{x} - \mathbf{x}_I)$ are their $(r - l + 1)$ th order approximations, respectively. In fact, when $m = r$, there is no new coefficient which has been computed in step 1(a). However, the following argument will show that the new values computed in step 1(f) for coefficients of the r th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$ are their first order approximations, respectively.

Let $L_I(\mathbf{x} - \mathbf{x}_I) = c_0 + c_1 \cdot (\mathbf{x} - \mathbf{x}_I)$ in step 1(a) and let $\hat{L}(\mathbf{x} - \mathbf{x}_I) = \hat{c}_0 + \hat{c}_1 \cdot (\mathbf{x} - \mathbf{x}_I)$ be the corresponding linear part in the Taylor expansion of the same (as for U_J) $(m - 1)$ th partial derivative of $u(\mathbf{x})$ at $\mathbf{x}_I$. Therefore c_0 and c_1 approximate $\hat{c}_0$ and $\hat{c}_1$ to the order of $\mathcal{O}((\Delta x)^{r-m+2})$ and $\mathcal{O}((\Delta x)^{r-m+1})$, respectively. Also from the above assumptions it is easy to see that

$$\begin{aligned} \bar{R}_I &= (1/|C_I|) \int_{C_I} R_I(\mathbf{x} - \mathbf{x}_I) d\mathbf{x} + \mathcal{O}((\Delta x)^{r-m+2}), \\ \bar{R}_J &= (1/|C_J|) \int_{C_J} Q_I(\mathbf{x} - \mathbf{x}_I) d\mathbf{x} + \mathcal{O}((\Delta x)^{r-m+2}) \quad \text{for } J \neq I, \end{aligned}$$

$$\begin{aligned}\overline{\partial^{m-1}U_I} &= (1/|C_I|) \int_{C_I} [L_I(\mathbf{x} - \mathbf{x}_I) + R_I(\mathbf{x} - \mathbf{x}_I)] d\mathbf{x} \quad \text{and} \\ \overline{\partial^{m-1}U_J} &= (1/|C_J|) \int_{C_J} [L_J(\mathbf{x} - \mathbf{x}_I) + Q_J(\mathbf{x} - \mathbf{x}_I)] d\mathbf{x} + \mathcal{O}((\Delta x)^{r-m+2}) \quad \text{for } J \neq I.\end{aligned}\quad (3.1)$$

Therefore $\bar{L}_J = \overline{\partial^{m-1}U_J} - \bar{R}_J$ in step 1(d) approximates the cell average of $\widehat{L}(\mathbf{x})$ on cell C_J to the order of $\mathcal{O}((\Delta x)^{r-m+2})$, for all J .

Reconstructing $\tilde{L}_I(\mathbf{x} - \mathbf{x}_I) = \tilde{c}_0 + \tilde{c}_1 \cdot (\mathbf{x} - \mathbf{x}_I)$ from a stencil $C_{J_0}, C_{J_1}, \dots, C_{J_d} \in \{C_J\}$ is to find $\tilde{c}_0$ and $\tilde{c}_1$ satisfying the following linear system (see (2.1)):

$$\begin{aligned}\frac{1}{|C_{J_l}|} \int_{C_{J_l}} (\tilde{c}_0 + \tilde{c}_1 \cdot (\mathbf{x} - \mathbf{x}_I)) d\mathbf{x} &= \tilde{c}_0 + \tilde{c}_1 \cdot (\mathbf{x}_{J_l} - \mathbf{x}_I) \\ &= \bar{L}_{J_l} = \widehat{c}_0 + \widehat{c}_1 \cdot (\mathbf{x}_{J_l} - \mathbf{x}_I) + \mathcal{O}((\Delta x)^{r-m+2}),\end{aligned}\quad (3.2)$$

where $\mathbf{x}_{J_l}$ is the cell centroid of cell C_{J_l} , $l = 0, \dots, d$. The solutions are candidates for c_0 and c_1 , respectively. Subtracting, e.g., the first equation ($l = 0$) from the rest of the equations in (3.2) we can obtain

$$A^T(\tilde{c}_1 - \widehat{c}_1) = \mathcal{O}((\Delta x)^{r-m+1}),$$

where $A = (1/\Delta x)[\mathbf{x}_{J_1} - \mathbf{x}_{J_0}, \mathbf{x}_{J_2} - \mathbf{x}_{J_0}, \dots, \mathbf{x}_{J_d} - \mathbf{x}_{J_0}]$. From condition 1, $\|A^{-1}\|$ is bounded independent of Δx . We conclude that the candidate

$$\tilde{c}_1 = \widehat{c}_1 + \mathcal{O}((\Delta x)^{r-m+1}). \quad (3.3)$$

Since the function F used in step 1(f) is a convex combination of its arguments, it does not change the approximation order of its arguments. Estimate (3.3) does not involve any new coefficient when $m = r$, thus this shows that the new values computed in step 1(f) for the coefficients of the r th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$ are their first order approximations, respectively. Therefore by induction for $m = r - 1, \dots, 1$, estimate (3.3) implies that the new values for coefficients of the m th degree terms of $U_I(\mathbf{x} - \mathbf{x}_I)$ are their $(r - m + 1)$ th order approximations. Along with step 2, we conclude that $\tilde{U}_I(\mathbf{x} - \mathbf{x}_I) = u(\mathbf{x}) + \mathcal{O}((\Delta x)^{r+1})$ for $\mathbf{x} \in C_I$, and the cell average of $U_I(\mathbf{x} - \mathbf{x}_I)$ on cell C_I is unchanged with the new coefficients. The proof is now complete.

4. Application to 1D conservation laws

We now use the finite volume scheme with HR to solve the 1D conservation law:

$$\begin{aligned}\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} &= 0, \quad (x, t) \in R \times (0, T), \\ u(x, 0) &= u_0(x), \quad x \in R.\end{aligned}\quad (4.1)$$

Let $\{x_i\}$ be a uniform partition in R with $\Delta x = x_{i+1} - x_i$ and $x_{i+1/2} = \frac{1}{2}(x_i + x_{i+1})$. The interval $(x_{i-1/2}, x_{i+1/2})$ is considered as a cell for all i . Integrating (4.1) over $(x_{i-1/2}, x_{i+1/2})$ we obtain

$$\frac{d}{dt} \bar{u}_i = -\frac{1}{\Delta x} \{f(u)|_{x_{i+1/2}} - f(u)|_{x_{i-1/2}}\},$$

where $\bar{u}_i$ is the cell average of u over $(x_{i-1/2}, x_{i+1/2})$. Let $\bar{U}_i$ be the numerical cell average approximating $\bar{u}_i$. Let $\tilde{U}_i(x)$ be a polynomial defined on $(x_{i-1/2}, x_{i+1/2})$ for all i , reconstructed out of $\{\bar{U}_i\}$. This reconstruction procedure is a major step for a high order finite volume scheme

$$\frac{d}{dt} \bar{U}_i = -\frac{1}{\Delta x} \{F_{i+1/2} - F_{i-1/2}\},$$

where $F_{i+1/2} = h(\tilde{U}_i(x_{i+1/2}-), \tilde{U}_{i+1}(x_{i+1/2}+))$ is a flux function. Here we use the Lax–Friedrich flux function $h(a, b) = \frac{1}{2}[f(a) + f(b)] + \frac{\alpha}{2}(a - b)$, where $\alpha = \max_u |f'(u)|$ is the largest characteristic speed, see [22] for more details. We use the third order TVD Runge–Kutta method [23] for time discretization. For systems of conservation laws, the reconstruction is applied to conservative variables (component-wise) without using local characteristic decomposition.

4.1. Preliminary reconstruction

Given the numerical cell averages $\{\bar{U}_i\}$ at a time t , we use a central reconstruction to form a piecewise polynomial function $U_i(x)$, $x \in (x_{i-1/2}, x_{i+1/2})$ for all i , although the reconstructed function may contain spurious oscillations in non-smooth regions of the solution (see e.g. [16]).

Fourth order case. The reconstructed function is a piecewise third degree polynomial ($r = 3$). For convenience, rewrite the reconstructed third degree polynomial in $(x_{i-1/2}, x_{i+1/2})$ as $U_i(x - x_i) = c_0 + c_1(x - x_i) + c_2(x - x_i)^2 + c_3(x - x_i)^3$. The coefficients of this polynomial can be determined by a least-square fit [3, 10]:

$$\begin{aligned} & \text{Minimize} \quad \sum_{\{j: 0 < |j-i| \leq 2\}} \left\{ \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} U_i(x - x_i) dx - \bar{U}_j \right\}^2, \\ & \text{subject to} \quad \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} U_i(x - x_i) dx = \bar{U}_i. \end{aligned}$$

Fifth order case. The reconstructed function is a piecewise fourth degree polynomial ($r = 4$). For convenience, rewrite the reconstructed fourth degree polynomial in $(x_{i-1/2}, x_{i+1/2})$ as $U_i(x - x_i)$. The coefficients of this polynomial can be determined by solving the following linear system [1]:

$$\frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} U_i(x - x_i) dx = \bar{U}_j,$$

for all j such that $|j - i| \leq 2$.

4.2. Hierarchical reconstruction

By using a preliminary reconstruction we have obtained a polynomial solution of degree r , $U_i(x - x_i)$, $x \in (x_{i-1/2}, x_{i+1/2})$, for all i . We then apply HR (algorithm 2), to recompute each polynomial and remove possible spurious oscillations. This procedure can also be applied wherever necessary to reduce the cost (see, e.g., [16, 30]), which is out of our focus here. Suppose we want to recompute $U_i(x - x_i)$ in $(x_{i-1/2}, x_{i+1/2})$ to obtain $\tilde{U}_i(x - x_i)$. Using the partial neighbouring cell idea [30] and notation of algorithm 2, we set

$$\begin{aligned} C_I &= (x_{i-1/2}, x_{i+1/2}), \\ \{C_J\} &= \{(x_{i-1}, x_{i-1/2}), C_I, (x_{i+1/2}, x_{i+1})\}, \end{aligned}$$

where $(x_{i-1}, x_{i-1/2})$ and $(x_{i+1/2}, x_{i+1})$ are partial neighbouring cells where the supported polynomials are still $U_{i-1}(x - x_{i-1})$ and $U_{i+1}(x - x_{i+1})$, respectively. The only two stencils

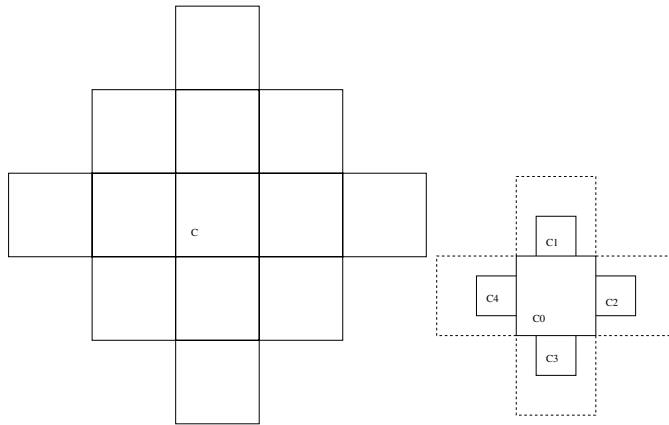


Figure 1. Left: the preliminary reconstruction on cell C depends on 13 nearby cell averages. Right: HR (algorithm 2) for cell C_0 uses 4 partial neighbouring cells C_1, C_2, C_3 and C_4 .

used in step 1(e) of algorithm 2 are $\{(x_{i-1}, x_{i-1/2}), C_I\}$ and $\{C_I, (x_{i+1/2}, x_{i+1})\}$. The limiter function F of algorithm 2 is taken to be the centre biased minmod function [26]

$$F(a, b) = \begin{cases} \min\{(1 + \epsilon)a, (1 + \epsilon)b, (a + b)/2\}, & \text{if } a, b > 0, \\ \max\{(1 + \epsilon)a, (1 + \epsilon)b, (a + b)/2\}, & \text{if } a, b < 0, \\ 0, & \text{otherwise,} \end{cases}$$

where ϵ is a small perturbation parameter taken to be 0.01 [16] in this paper (see [20, 21] for a discussion for the loss of accuracy due to abrupt shift of stencils, which is the reason for using a perturbation ϵ here).

5. Application to 2D conservation laws

The fourth order finite volume implementation on a 2D rectangular grid is similar to the 1D case discussed in the previous section, see [16] for more details. The difference from [16] is that algorithm 2 on partial neighbouring cells (instead of algorithm 1) is used. The preliminary reconstruction of a third degree polynomial of two variables on a cell C (left of figure 1) is based on a least-square fit over 13 nearby cell averages. In order to eliminate possible spurious oscillations introduced by the preliminary reconstruction, applying HR (algorithm 2) to a cell C_0 (right of figure 1) involves 4 partial neighbouring cells C_1, C_2, C_3 and C_4 with half of the original size. The polynomial defined on a partial neighbouring cell is the same as the one defined on the original cell containing this partial neighbouring cell. The stencils used in algorithm 2 are $\{C_0, C_1, C_2\}$, $\{C_0, C_2, C_3\}$, $\{C_0, C_3, C_4\}$ and $\{C_0, C_4, C_1\}$.

5.1. Numerical examples

Example 1. We test algorithm 2 with partial neighbouring cells by using the Burgers equation $u_t + (\frac{1}{2}u^2)_x = 0$, $u(x, 0) = \frac{1}{4} + \frac{1}{2}\sin(\pi x)$. The errors are shown in tables 1 and 2 at the time $T = 0.1$ when the solution is still smooth. In order to match the spatial order of accuracy, the time step size for the fourth order case is chosen with CFL number 0.9 or as $\Delta x^{4/3}$, whichever is smaller; the time step size for the fifth order case is chosen with CFL number 0.9 or as $\Delta x^{5/3}$, whichever is smaller. Besides the current time step sizes,

Table 1. Convergence test for the Burgers equation: fourth order case; algorithm 2 with partial neighbouring cells.

Δx	1/10	1/20	1/40	1/80	1/160	1/320	1/640
L_1 error	1.10E-4	2.65E-6	1.01E-7	5.31E-9	3.06E-10	1.90E-11	1.18E-12
Order	—	5.38	4.71	4.25	4.12	4.01	4.01
L_∞ error	1.96E-4	7.48E-6	2.27E-7	1.65E-8	9.99E-10	5.69E-11	3.34E-12
Order	—	4.71	5.04	3.78	4.05	4.13	4.09

Table 2. Convergence test for the Burgers equation: fifth order case; algorithm 2 with partial neighbouring cells.

Δx	1/10	1/20	1/40	1/80	1/160	1/320
L_1 error	2.49E-5	1.60E-7	7.14E-9	2.92E-10	1.02E-11	3.37E-13
Order	—	7.28	4.49	4.61	4.84	4.92
L_∞ error	7.23E-5	8.65E-7	1.40E-08	5.95E-10	2.71E-11	1.41E-12
Order	—	6.39	5.95	4.56	4.46	4.26

Table 3. Convergence test for the Burgers equation: fifth order case; algorithm 2 with partial neighbouring cells in which mixed limiter functions are used.

Δx	1/10	1/20	1/40	1/80	1/160	1/320
L_1 error	1.99E-5	1.47E-7	6.95E-9	2.87E-10	9.94E-12	3.25E-13
Order	—	7.08	4.40	4.60	4.85	4.93
L_∞ error	6.00E-5	8.66E-7	1.29E-8	5.59E-10	1.97E-11	6.59E-13
Order	—	6.11	6.07	4.53	4.83	4.90

we have also tried smaller time step sizes and found no significant difference in the error. The convergence rates roughly match the expected orders of convergence at least in the l_1 norm. The order of the l_∞ error for the fifth order case (in table 2) gradually reduces to 4 as we refine the mesh. This can be overcome by replacing the limiter function $F(a, b)$ used for recomputing the fourth and third degree coefficients in algorithm 2 by a smoother one described as follows which uses a WENO-type weighted average as in [16]. $F(a, b) = \theta_a a + \theta_b b$, if $ab > 0$; $F(a, b) = 0$ otherwise. Here $\theta_a = [1/(1 + \Delta x a^4)]/s$, $\theta_b = [1/(1 + \Delta x b^4)]/s$, and $s = [1/(1 + \Delta x a^4)] + [1/(1 + \Delta x b^4)]$. This ensures that if $a \in (\mathcal{O}(\Delta x^{-1}), \mathcal{O}(\Delta x^{-4}))$, $b = \mathcal{O}(1)$ and $ab > 0$, then $\theta_a a = \mathcal{O}(\Delta x^2)$ which is within the approximation error order of the fourth and third degree coefficients. We call them the mixed limiter functions and use them to obtain better l_∞ error in table 3. In fact, the errors in table 3 are very close to those of table 4 in which the limiter functions are all set to be $F(a, b) = \frac{1}{2}(a + b)$ without any limiting.

Example 2. Consider the two-dimensional Burgers equation with periodic boundary conditions:

$$\begin{aligned} \partial_t u + \partial_x \left(\frac{u^2}{2} \right) + \partial_y \left(\frac{u^2}{2} \right) &= 0, \quad \text{in } (0, T) \times \Omega, \\ u(t = 0, x, y) &= \frac{1}{4} + \frac{1}{2} \sin(\pi(x + y)), \quad (x, y) \in \Omega, \end{aligned} \quad (5.1)$$

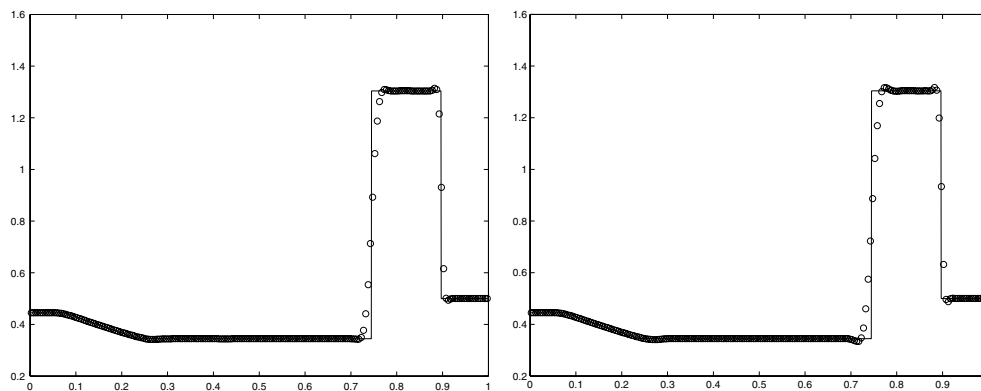
where $\Omega = [-1, 1] \times [-1, 1]$. At $T = 0.1$ the exact solution is smooth. We test the fourth order finite volume scheme with algorithm 2 and partial neighbouring cells (section 5) on rectangular grids. The time step is taken with CFL number 0.5 or as $\Delta x^{4/3}$, whichever is

Table 4. Convergence test for the Burgers equation: fifth order case; algorithm 2 with partial neighbouring cells in which the limiter function is replaced by the average of its arguments.

Δx	1/10	1/20	1/40	1/80	1/160	1/320
L_1 error	1.16E-5	9.61E-8	6.47E-9	2.74E-10	9.66E-12	3.22E-13
Order	—	6.92	3.89	4.56	4.83	4.91
L_∞ error	2.43E-5	1.57E-7	1.27E-8	5.55E-10	1.96E-11	6.56E-13
Order	—	7.27	3.63	4.52	4.82	4.90

Table 5. Convergence test for the 2D Burgers equation: fourth order case; algorithm 2 with partial neighbouring cells.

Δx	1/8	1/16	1/32	1/64	1/128
L_1 error	3.03E-3	1.65E-4	7.89E-6	3.87E-7	2.22E-8
Order	—	4.20	4.39	4.35	4.12
L_∞ error	3.22E-3	2.47E-4	1.08E-5	6.05E-7	4.03E-8
Order	—	3.70	4.52	4.16	3.91

**Figure 2.** Lax problem: $N = 200$, fourth order case; left: algorithm 2 with partial neighbouring cells, right: algorithm 1 without using partial neighbouring cells.

smaller. The errors at the time $T = 0.1$ are listed in table 5, which clearly show the fourth order convergence rate.

Example 3. We compute the Euler equations with Lax's initial data [12]. $u_t + f(u)_x = 0$ with $u = (\rho, \rho v, E)^T$, $f(u) = (\rho v, \rho v^2 + p, v(E + p))^T$, $p = (\gamma - 1)(E - \frac{1}{2}\rho v^2)$ and $\gamma = 1.4$. Initially, the density ρ , momentum ρv and total energy E are 0.445, 0.311 and 8.928 in $(0, 0.5)$; and are 0.5, 0 and 1.4275 in $(0.5, 1)$. With the CFL number equal to 0.9 and $\Delta x = 1/200$, the computed results are shown at the time $T = 0.16$ in figures 2, 3 and on the left of figure 6. The solid line reference solution is the analytic solution to the Riemann problem. We observe that overshoots/undershoots are essentially eliminated with algorithm 2 on partial neighbouring cells. Also the two different limiter functions used in the fifth order case make almost no difference in the graphs.

Example 4 (Shu–Osher problem [24]). It is the Euler equations with an initial data

$$\begin{aligned}
 (\rho, v, p) &= (3.857\,143, 2.629\,369, 10.333\,333), & \text{for } x < -4, \\
 (\rho, v, p) &= (1 + 0.2 \sin(5x), 0, 1), & \text{for } x \geq -4.
 \end{aligned}$$

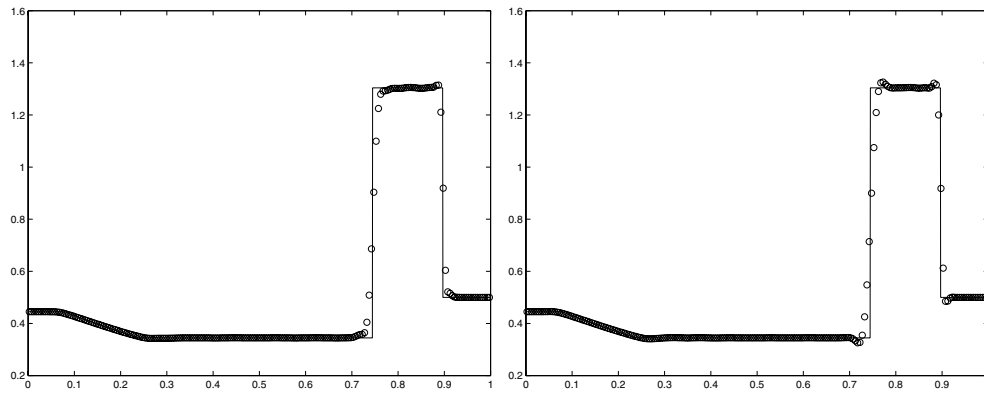


Figure 3. Lax problem: $N = 200$, fifth order case; left: algorithm 2 with partial neighbouring cells, right: algorithm 1 without using partial neighbouring cells.

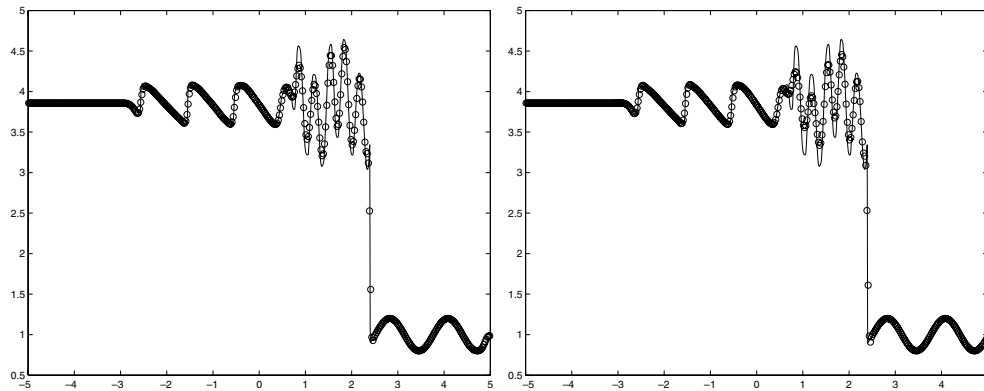


Figure 4. Shu–Osher problem: $N = 400$, fourth order case; left: algorithm 2 with partial neighbouring cells, right: algorithm 1 without using partial neighbouring cells.

The density profiles are plotted at the time $T = 1.8$ in figures 4, 5 and the right of figure 6, with $\Delta x = 1/40$ and the CFL number equal to 0.9. The solid line is the numerical solution on a fine mesh ($\Delta x = 1/200$) computed by a central scheme on overlapping cells [14]. We observe better resolution with algorithm 2 on partial neighbouring cells. Also the two different limiter functions used in the fifth order case make almost no difference in the graphs.

Example 5. We compute the 2D Euler equations with the initial data of Lax (example 3), or Shu and Osher (example 4) set along the x -direction, constant initial data along any y -direction, and 0 y -velocity. The 2D Euler equations can be written as

$$\mathbf{u}_t + \mathbf{f}(\mathbf{u})_x + \mathbf{g}(\mathbf{u})_y = 0, \quad \mathbf{u} = (\rho, \rho u, \rho v, E)^T, \quad p = (\gamma - 1)(E - \frac{1}{2}\rho(u^2 + v^2))$$

$$\mathbf{f}(\mathbf{u}) = (\rho u, \rho u^2 + p, \rho uv, u(E + p))^T, \quad \mathbf{g}(\mathbf{u}) = (\rho v, \rho uv, \rho v^2 + p, v(E + p))^T,$$

where $\gamma = 1.4$. The density profiles along the x -direction are plotted in figure 7, computed by the fourth order finite volume scheme on rectangular grids with algorithm 2 and partial neighbouring cells (section 5). The CFL number is equal to 0.5. Clearly, the graphs are similar to their 1D versions on the left of figure 2 and on the left of figure 4.

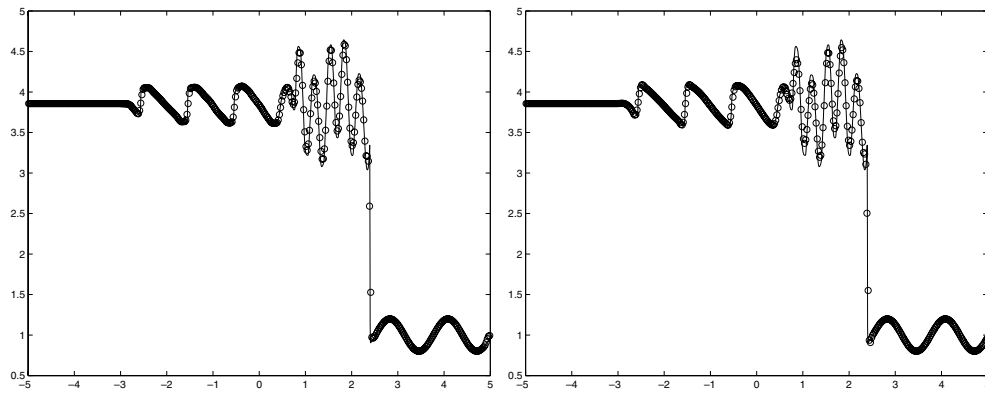


Figure 5. Shu–Osher problem: $N = 400$, fifth order case; left: algorithm 2 with partial neighbouring cells, right: algorithm 1 without using partial neighbouring cells.

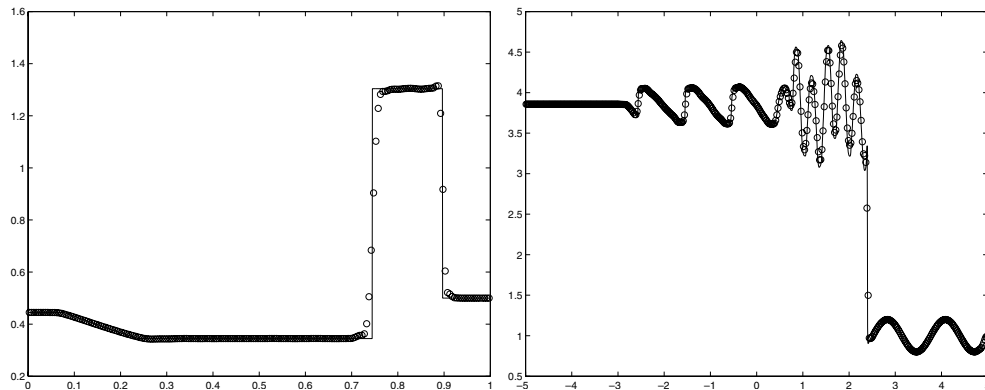


Figure 6. Algorithm 2 with partial neighbouring cells: fifth order case with mixed limiter functions; left: Lax problem, $N = 200$, right: Shu–Osher problem, $N = 400$.

Example 6. We compute the double Mach reflection [29] by the fourth order finite volume scheme using algorithm 2 and partial neighbouring cells (section 5). It is governed by the 2D Euler equations with the following initial data. A planar Mach 10 shock is incident on an oblique wedge at a $\pi/3$ angle. The air in front of the shock has density 1.4, pressure 1 and velocity 0. The boundary conditions are described in [29].

The CFL number is equal to 0.5. The density profile is plotted at $T = 0.2$ at the top of figure 8 with 30 equally spaced contours. The comparative result with algorithm 1 [16] is plotted at the bottom of figure 8. It is clear that the top graph shows more rolling in the Mach stem. A fix of negative pressure is necessary for this test problem in [16] with algorithm 1, which is due to the high pressure ratio across strong shocks and overshoots/undershoots. However, using algorithm 2 on partial neighbouring cells here, we never detect any negative pressure, which is clearly attributed to the reduction of overshoots/undershoots. In all simulations of this example, HR is only applied in the non-smooth region as in [16] in order to reduce the computational cost.

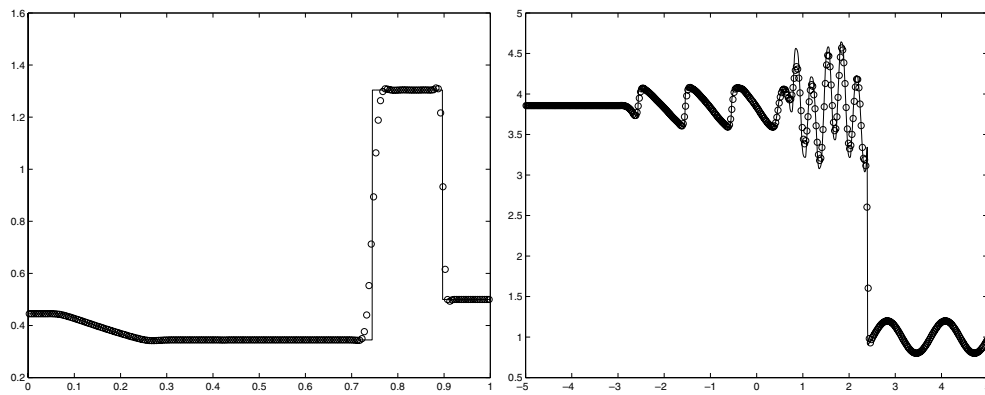


Figure 7. Algorithm 2 with partial neighbouring cells on 2D rectangular grid: fourth order case, density profile along the x -direction; left: Lax problem, $\Delta x = \Delta y = 1/200$, right: Shu–Osher problem, $\Delta x = \Delta y = 1/40$.

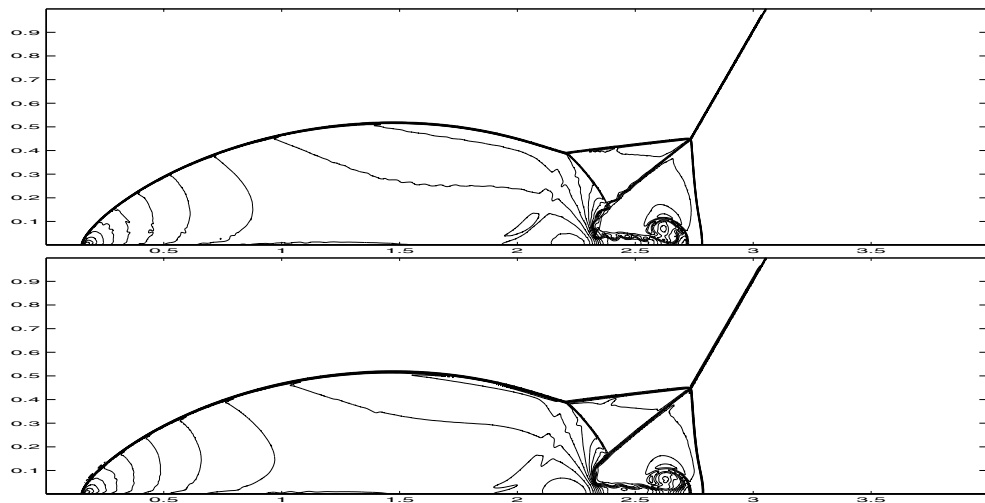


Figure 8. Density contour of double Mach reflection: fourth order case, $\Delta x = \Delta y = 1/480$; top: algorithm 2 with partial neighbouring cells, bottom: algorithm 1.

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References

- [1] Abgrall R 1994 On essentially non-oscillatory schemes on unstructured meshes: analysis and implementation *J. Comput. Phys.* **114** 45–58
- [2] Biswas R, Devine K and Flaherty J 1994 Parallel, adaptive finite element methods for conservation laws *Appl. Numer. Math.* **14** 255–83

- [3] Barth T and Frederickson P 1990 High order solution of the Euler equations on unstructured grids using quadratic reconstruction *AIAA Paper* No 90-0013
- [4] Cockburn B and Shu C-W 1998 The TVB Runge–Kutta local projection discontinuous Galerkin finite element method for conservation laws: V. Multidimensional systems *J. Comput. Phys.* **141** 199–224
- [5] Cockburn B, Lin S-Y and Shu C-W 1989 TVB Runge–Kutta local projection discontinuous Galerkin finite element method for conservation laws: III. One dimensional systems *J. Comput. Phys.* **84** 90–113
- [6] Cockburn B and Shu C-W 1989 TVB Runge–Kutta local projection discontinuous Galerkin finite element method for conservation laws: II. General framework *Math. Comput.* **52** 411–35
- [7] Godunov S K 1959 *Mat. Sb.* **47** 357
- [8] Harten A 1983 High resolution scheme for hyperbolic conservation laws *J. Comput. Phys.* **49** 357–93
- [9] Harten A, Engquist B, Osher S and Chakravarthy S 1987 Uniformly high order accurate essentially non-oscillatory schemes: III *J. Comput. Phys.* **71** 231–303
- [10] Hu C-Q and Shu C-W 1999 Weighted essentially non-oscillatory schemes on triangular meshes *J. Comput. Phys.* **150** 97–127
- [11] Jiang G-S and Shu C-W 1996 Efficient implementation of weighted ENO schemes *J. Comput. Phys.* **126** 202–28
- [12] Lax P 1954 Weak solutions of nonlinear hyperbolic equations and their numerical computations *Commun. Pure Appl. Math.* **7** 159
- [13] Liu X-D, Osher S and Chan T 1994 Weighted essentially non-oscillatory schemes *J. Comput. Phys.* **115** 200–12
- [14] Liu Y-J 2005 Central schemes on overlapping cells *J. Comput. Phys.* **209** 82–104
- [15] Liu Y-J, Shu C-W, Tadmor E and Zhang M-P 2007 Central discontinuous Galerkin methods on overlapping cells with a non-oscillatory hierarchical reconstruction *SIAM J. Numer. Anal.* **45** 2442–67
- [16] Liu Y-J, Shu C-W, Tadmor E and Zhang M-P 2007 Non-oscillatory hierarchical reconstruction for central and finite volume schemes *Commun. Comput. Phys.* **2** 933–63
- [17] Qiu J and Shu C-W 2002 On the construction, comparison, and local characteristic decomposition for high-order central WENO schemes *J. Comput. Phys.* **183** 187–209
- [18] Qiu J and Shu C-W 2005 Runge–Kutta discontinuous Galerkin method using WENO limiters *SIAM J. Sci. Comput.* **26** 907–29
- [19] Reed W and Hill T 1973 Triangular mesh methods for the neutron transport equation *Technical Report* la-ur-73-479, Los Alamos Scientific Laboratory
- [20] Rogerson A M and Meiburg E 1990 A numerical study of the convergence properties of ENO schemes *J. Sci. Comput.* **5** 151–67
- [21] Shu C-W 1990 Numerical experiments on the accuracy of ENO and modified ENO schemes *J. Sci. Comput.* **5** 127–49
- [22] Shu C-W 1998 Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws *Advanced Numerical Approximation of Nonlinear Hyperbolic Equations* ed B Cockburn et al (*Lecture Notes in Mathematics*) ed A Quarneroni (Berlin: Springer) p 1697
- [23] Shu C-W and Osher S 1988 Efficient implementation of essentially non-oscillatory shock capturing schemes *J. Comput. Phys.* **77** 439–71
- [24] Shu C-W and Osher S 1989 Efficient implementation of essentially non-oscillatory shock capturing schemes: II *J. Comput. Phys.* **83** 32–78
- [25] van Leer B 1974 Toward the ultimate conservative difference scheme: II. Monotonicity and conservation combined in a second order scheme *J. Comput. Phys.* **14** 361–70
- [26] van Leer B 1977 Towards the ultimate conservative difference scheme: IV. A new approach to numerical convection *J. Comput. Phys.* **23** 276–99
- [27] van Leer B 1979 Towards the ultimate conservative difference scheme: V. A second order sequel to Godunov's method *J. Comput. Phys.* **32** 101–36
- [28] Wang Z J and Liu Y 2004 Spectral (finite) volume method for conservation laws on unstructured grids: III. Extension to one-dimensional systems *J. Sci. Comput.* **20** 137–57
- [29] Woodward P and Colella P 1984 Numerical simulation of two-dimensional fluid flows with strong shocks *J. Comput. Phys.* **54** 115
- [30] Xu Z-L, Liu Y-J and Shu C-W 2009 Hierarchical reconstruction for discontinuous Galerkin methods on unstructured grids with a WENO type linear reconstruction and partial neighboring cells *J. Comput. Phys.* **228** 2194–212
- [31] Xu Z-L, Liu Y-J and Shu C-W 2009 Hierarchical reconstruction for spectral volume method on unstructured grids *J. Comput. Phys.* **228** 5787–802